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Approach:

1. Set ans = 0 and mod = $10^9 + 7$
2. Let n be array size
3. For each bit from 0 to 30
   - count how many numbers have this bit = 1 → count1.
   - count0 = n - count1.
4. Pairs differing at this bit count1 * count0.
   Add contribution ans = ans + (2 * count1 * count0) (mod)
5. Return ans.

Name - Aditya Singh
UID- 23BCS11827
Section - KRG-2B

Test Cases

Input 1:
A= [1,3,5]

Output 1:
8

Input 2:
A= [2,3]

Output 2:
2

```
public class Solution {
    public int solve(int[] A) {
        long mod = 1000000007;
        int n = A.length;
        long ans = 0;
        for(int bit=0 ; bit < 31 ; bit++){
            long count1 = 0;
            for(int i = 0 ; i<n ; i++){
                if((A[i] & (1<<bit)) != 0){
                    count1++;
                }
            }
            long count0 = n- count1;
            ans= (ans + (2* count1 * count0)%mod)%
                mod;
        }
        return (int) ans;
    }
}
```